Name_____ Date_____ Period_____

LESSON 6-1 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Dividing by a Unit Fraction

Maryland Standard 6.NS.A.1 · Learning target – I can divide a whole number by a unit fraction and explain what the quotient counts.

REMEMBER

- Dividing by a **unit fraction** asks: how many pieces of that size fit into the number?
- 1 whole holds 4 fourths, so **3 ÷ 1/4 = 3 × 4 = 12**.
- The quotient is **greater** than the number you divide. Check: 12 × 1/4 = 3.

WORKED EXAMPLE

How many 1/3-cup scoops are in 5 cups of rice?

Step 1 Each cup holds 3 thirds.

Step 2 5 cups hold 5 × 3 = 15 thirds, so 5 ÷ 1/3 = 15.

Step 3 Check: 15 × 1/3 = 5, so it checks.

Answer **15 scoops**

WORD BANK

unit fraction – a fraction with 1 on top, like 1/4

quotient – the answer when you divide

reciprocal – the flipped number: 1/4 becomes 4

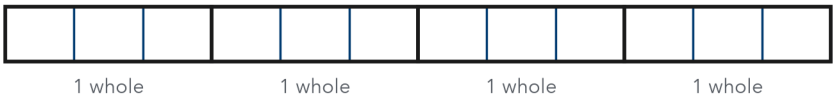
PRACTICE

show your work · label units

1. Each bar is 1 whole. Use the fraction bars to find **4 ÷ 1/3**. How many thirds are in 4 wholes?

Step 1 Count the thirds in 1 bar.

Step 2 Count the thirds in all 4 bars.



4 ÷ 1/3 =_____

2. Divide.

a. 6 ÷ 1/2 = _____ **b.** 3 ÷ 1/5 = _____ **c.** 8 ÷ 1/4 = _____

d. 2 ÷ 1/9 = _____ **e.** 7 ÷ 1/3 = _____ **f.** 10 ÷ 1/6 = _____

g. 5 ÷ 1/7 = _____ **h.** 12 ÷ 1/3 = _____

3. **Trail Mix.** Jamal has **6 cups** of trail mix. Each snack bag holds **1/4 cup**. How many bags can he fill?

   Step 1 Write the division: 6 ÷ 1/4.

   Step 2 Change it to multiplication: 6 × 4.

   Answer_____ bags

4. **Ribbon Bows.** Ms. Carter has **5 yards** of ribbon. She cuts it into pieces that are **1/6 yard** long. She **uses 18** pieces. How many pieces are **left**?

   Step 1 First find 5 ÷ 1/6.

   Step 2 Then subtract the 18 pieces she uses.

   Show your work.

   Answer_____ pieces

5. **Find the Error.** Tanya says 8 ÷ 1/2 = 4. Explain her mistake. Then find the correct quotient.

   Frame *Tanya found ___ of 8. The correct answer is ___ because ___.*

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Count how many 1/4-cup scoops fill 2 cups. Does it match 2 ÷ 1/4?

Name_____ Date_____ Period_____

LESSON 6-1 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Dividing a Fraction by a Whole Number

Maryland Standard 6.NS.A.1 · Learning target – I can divide a fraction by a whole number and explain the result.

REMEMBER

- Dividing by a whole number **shares** a fraction into equal parts.
- Multiply by the reciprocal: **2/3 ÷ 4 = 2/3 × 1/4 = 2/12 = 1/6**.
- Dividing by a whole number greater than 1 gives a quotient **less** than the fraction. Check: 1/6 × 4 = 2/3.

WORKED EXAMPLE

3 friends share 3/4 of a sandwich equally. What part of the whole does each get?

Step 1 3/4 ÷ 3 = 3/4 × 1/3 = 3/12

Step 2 Simplify: 3/12 = 1/4

Answer **1/4 of the sandwich**

WORD BANK

share – split into equal parts

reciprocal – the flipped number: 6 becomes 1/6

simplest form – divide top and bottom by the same number

PRACTICE

show your work · label units

1. Divide. Write each quotient in simplest form.

a. 1/2 ÷ 3 = _____ **b.** 4/5 ÷ 2 = _____ **c.** 3/8 ÷ 3 = _____

d. 2/3 ÷ 6 = _____ **e.** 6/7 ÷ 2 = _____ **f.** 5/6 ÷ 10 = _____

2. Find each share. Write it in simplest form.

a. 4 people share 1/2 lb of cheese: ____ lb **b.** 5 cups share 5/6 L of juice: ____ L

3. Paint Cans. Kiara has **3/5 gallon** of paint. She pours it **equally** into **6 cans**. How much paint is in each can?

Step 1 Write the division: 3/5 ÷ 6.

Step 2 Multiply 3/5 by 1/6.

Step 3 Simplify your answer.

Show your work.

Answer_____ gallon

4. Clay. Mr. Okafor has 2/3 pound of clay. He splits it equally among 4 students. How much clay does each student get?

Step 1 Write the division: 2/3 ÷ 4.

Step 2 Multiply 2/3 by 1/4, then simplify.

Answer_____ pound

5. Reason. Without dividing, decide: is 5/8 ÷ 4 greater than or less than 5/8?

Frame *5/8 ÷ 4 is ___ than 5/8 because ___.*

☐ greater than 5/8 ☐ less than 5/8

TEST PRACTICE

choose the best answer

6. Which expression has the same value as 5/6 ÷ 5?

A 5/6 × 5 **B** 5/6 × 1/5 **C** 6/5 × 5 **D** 5/6 − 5

7. 4 people share **4/9** of a pizza **equally**. What fraction of the whole pizza does each person get?

A 1/9 **B** 16/9 **C** 1/36 **D** 4/9

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Share 1/2 of a sandwich equally among 2 people at home. What fraction of the whole sandwich does each get?